

Lab6 report

The main object of this lab is to build a pipelined CPU with fixed CBZ and Ubranch. The CPU pipeline can be broken into five stages. Thus, the compilation order is:

Firstly the basic components: AND2.vhd, add.vhd, OR2.vhd, MUX5.vhd, MUX64.vhd, MUX64_3.vhd.

And then the critical components of the CPU: alu.vhd, alucontrol.vhd, cpucontrol.vhd, PC.vhd, registers_p1.vhd, ShiftLeft2.vhd, SignExtend.vhd, lab5_dmem.chd, lab5_imem.vhd and cbz.

After that, we should compile the caches between the stages: IFID.vhd, IDEX.vhd, EXMEM.vhd, MEMWB.vhd.

Finally, the top-level design of the entire system: `piplinedcpu1.vhd` and my own testbench `piplinedcpu1_tb.vhd` (could be removed if you don't want it).

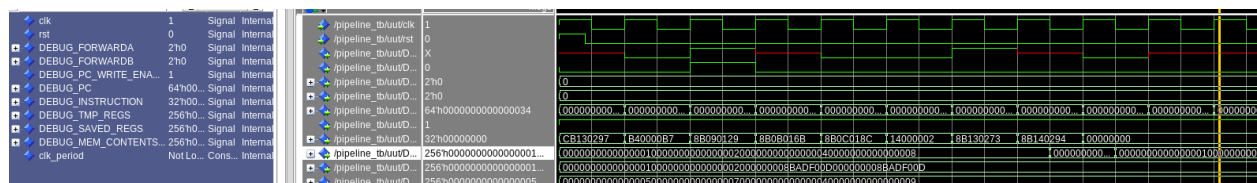
1. Instructions analysis

Using given instructions, we can draw the different stages of operating instructions:

CLK	1	2	3	4	5	6	7	8	9	10	11
SUB	IF	ID	EX	MEM	WB						
CBZ		IF	ID	flush	flush	flush	flush				
ADD			IF	flush	flush	flush	flush	flush			
ADD(7)				IF	ID	EX	MEM	WB			
ADD					IF	ID	EX	MEM	WB		
B2						IF	ID	flush	flush	flush	
ADD							IF	flush	flush	flush	Flush
ADD								IF	ID	EX	MEM

2.Signal Analysis

From the instruction analysis above, we can see that the third instruction is flushed because of the CBZ instruction before it. The full signal of the process is given as below:

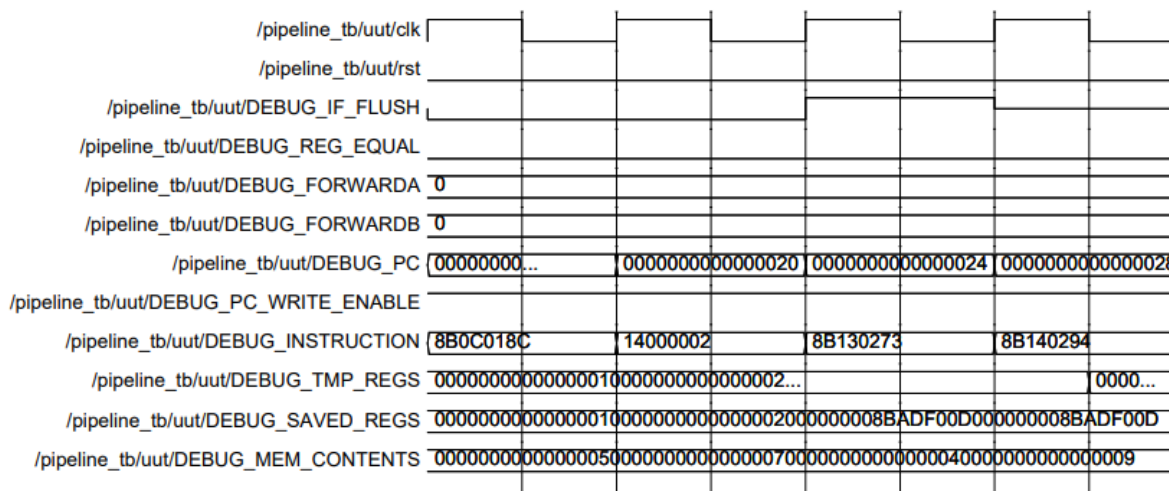
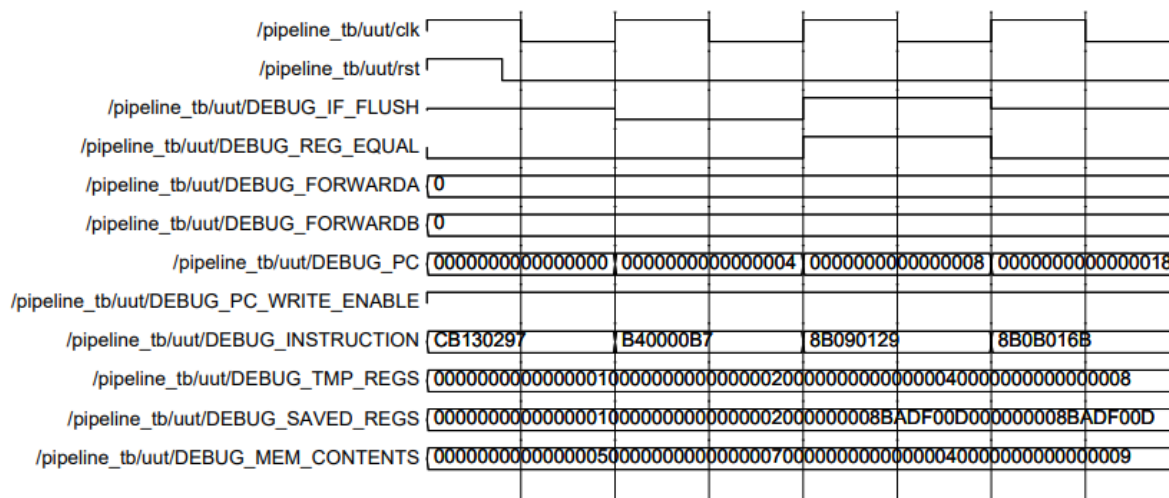


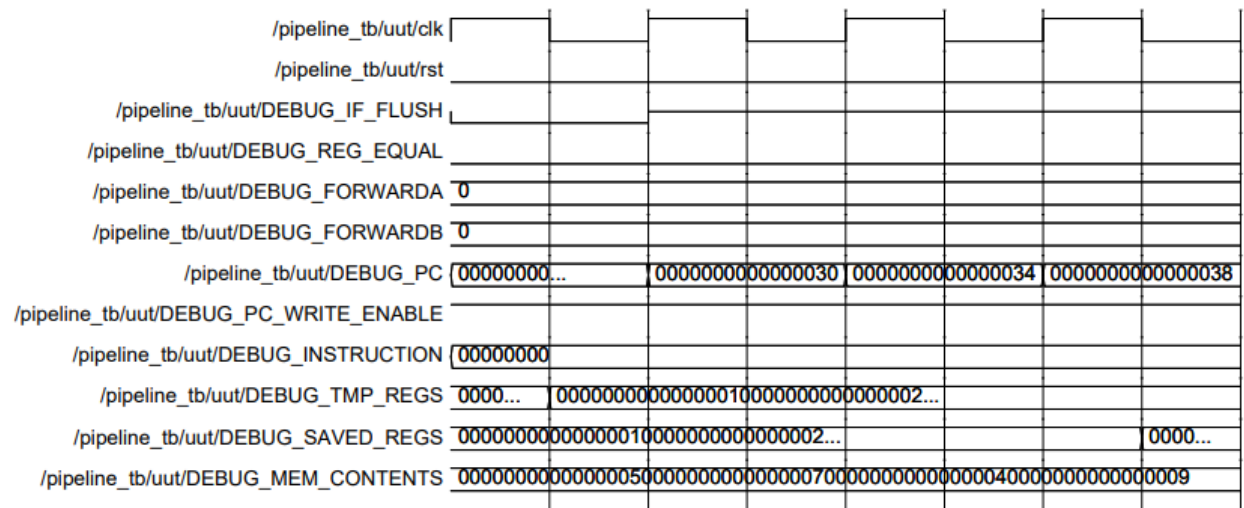
We can see that the output of the third signal, which should be displayed at 7th clock cycle, has been flushed. And the conditional branch has been executed at the third cycle, the unconditional branch is resolved at the 7th cycle.

Both predictions are incorrect because both branch are taken.

The flush occurs at the forth stage and seventh stage. That invalidate the effect of the instruction that is being fetched.

And also, because of the data hazard between “SUB X23, X20, X19” and “CBZ X23 5” the CPU dose jump to the 7th instruction. I will give out a signal figure more clearly.





By the way, I was also trying to forward the data from ALU result to the ID stage. Although theoretically it is possible, I failed either because of the unlimited flush or because of the non-flush at all. Thus, I finally give up that thought and make the pipelined CPU as required.